

Chasing the Nines

Reliable RL Across States and Seeds



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QUESTION: CAN ONE TRAINING ROUTE RELIABLY PRODUCE POLICIES ACROSS STATES AND SEEDS? · TRAINING COMPUTE DIFFERS.

Near-reference: pure RL 497 · mixed 9 failures; strict wins: pure 22.823% · mixed 12.555%

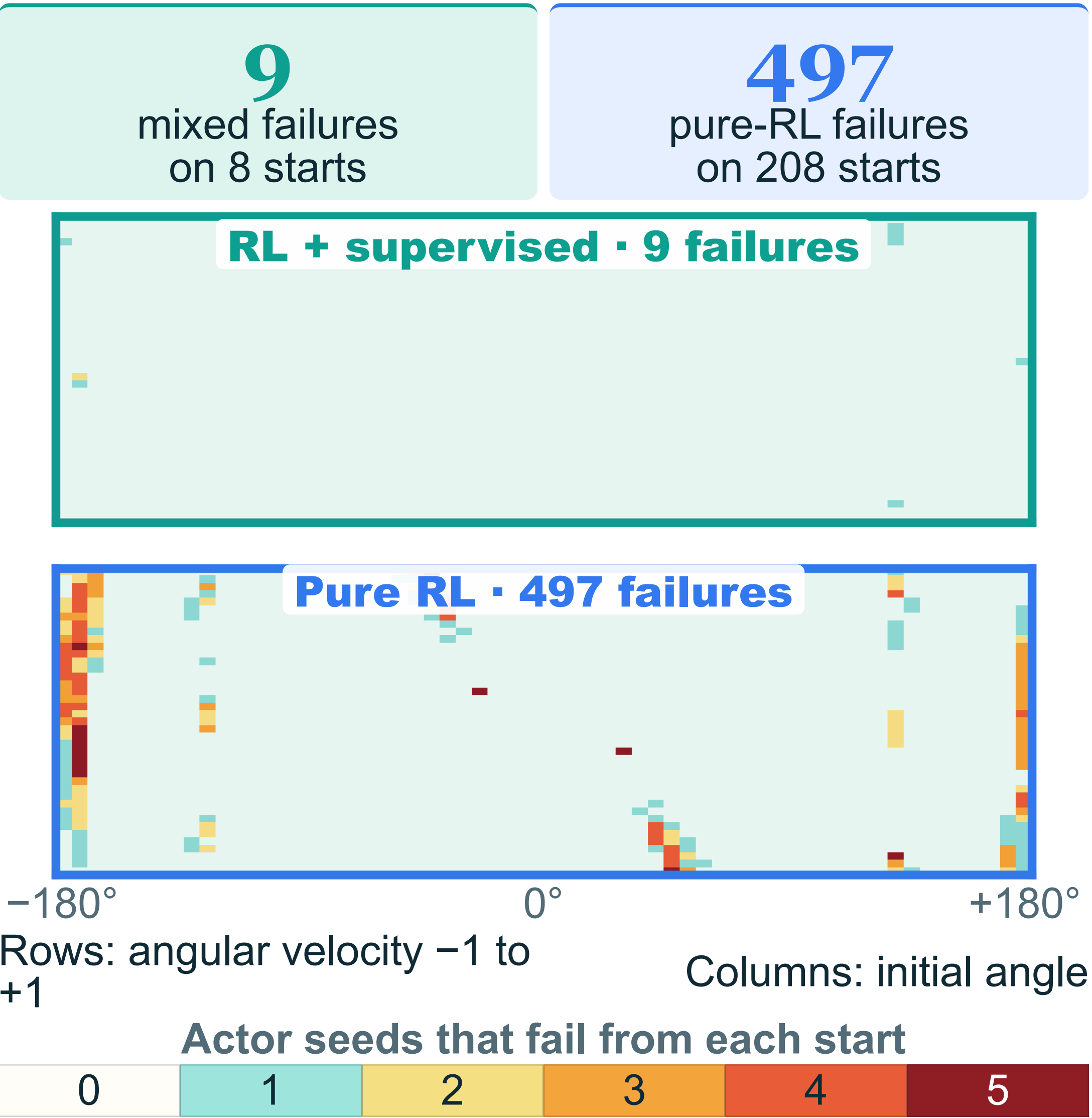
RL + supervised
9 / 12,505 failures
Task 93.858% · strict 12.555%

Near-reference
99.928%

Pure RL
497 / 12,505 failures
Task 92.875% · strict 22.823%

Near-reference
96.026%

01 Measure the reliability tail



All five mixed actors pass 2,493 of 2,501 starts. All five pure actors pass 2,293; another 195 depend on the training seed.

Fixed evaluation

Five actors × 61 evenly spaced angles × 41 velocities × 200 steps. Near: within 5 return of the better controller. Task: upright and slow for at least 80% of the rollout. Strict: beats both controllers.

Baselines: near / task / strict (%)

Supervised (1)	SimbaV2 (5)	Clean DAgger (5)
99.72 / 93.60 / 14.50	91.84 / 91.46 / 8.52	84.53 / 88.02 / 6.39

02 Build and test two policies

RL + supervised · sequential optimization updates: 30k DAgger + 20k priority DAgger + 50k critic

DAgger labels states visited by the learner. Priority training resamples the largest measured return deficits. Twin-Q tests actor action a_0 and offsets $\pm\{0.05, 0.10\}$; both critics must prefer a replacement.

Pure RL · 100k reward transitions

FastSACN8 averages one- and eight-step critic losses. Reflection uses $[\pi(s) - \pi(M(s))] / 2$ for mirrored state $M(s)$. Twin-Q tests 41 torques from −2 to 2; both predicted gains must exceed 0.005.

Three matched controls isolate the mixed route

38 without learner-state refit refit fixes 29	19 uniform start sampling priority fixes 10
35 without local Q-search search fixes 26	9 with the complete route five-seed result

Pure RL improves along one deployment path

1,267 raw actor failures	→	955 after reflection	→	497 after Q-search
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Mixed training collected 690k reference labels and 800k discovery interactions, so training compute differs. Deployment uses one policy pipeline and the current state only. Neither route queries a reference controller.

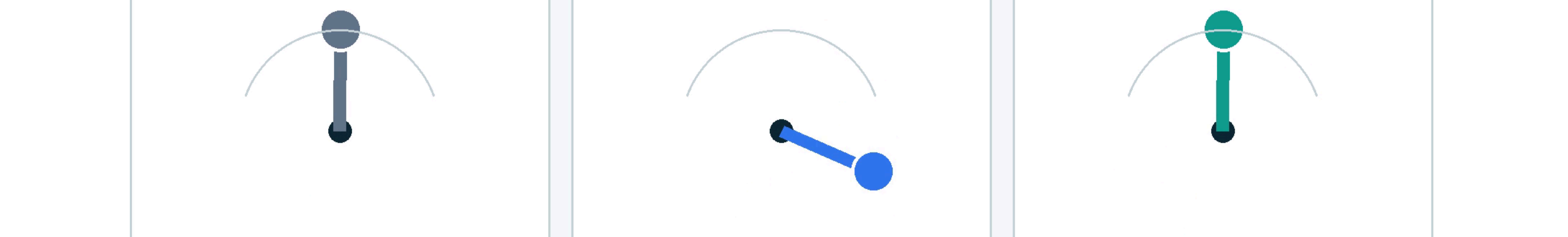
Can supervised and SACN8 losses train together?

2,443 of 2,501 starts joint BC + SACN8	1,757 of 2,501 starts reward-only SACN8	37.2× median weighted BC/SAC gradient norm ratio	−0.943 to +0.935 gradient cosine range
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One-seed, 25k-step pilot. Relative gradient scale and alignment both change during training, motivating normalized or projected mixing.

03 Why pure RL misses the tail

Diagnostic reference **Pure-RL raw actor** **RL + supervised actor**



Same start, step 81. Final return: reference −280.9; pure actor −384.3; RL + supervised −280.6.

Inside matched actors and critics

Five seeds, 100k transitions, the same SimbaV2 residual network, and one critic update per transition. SAC uses one step; FastSACN8 averages one and eight; SACN8 uses eight. A/B are medians of five seed values over 5,553 fixed states. C/D pool 2,560 fixed raw-actor failures, 512 per seed.

A · Actor output: SAC → FastSACN8

59% → 87%

States at the torque bound, SAC versus FastSACN8.

B · Tanh sensitivity: SAC → FastSACN8

0.120 → 0.039

Median derivative from actor logit to torque.

C · Frozen critic

75%

The action gradient points toward a higher-return correction.

D · Action clipping

7%

Median critic step retained after bounded-action projection.

The measurements are consistent with an actor-side bottleneck: critic directions are often useful, while saturation removes most of the local update. Scoring 41 torques bypasses projection and raises near-reference success from 92.363% to 96.026%.

RL plus supervision leaves **9 near-reference failures in 12,505 trials**. Pure RL reaches 96.026% near-reference success and 22.823% strict wins; its internal measurements are consistent with actor saturation and clipping in the remaining tail.

Reference controllers: dynamic programming and energy shaping
SAC: Haarnoja et al., 2018 · DAgger: Ross et al., 2011
SimbaV2: Lee et al., 2025 · SACn: Łyskawa et al., 2025